
(a) Full network $G$
(b) Two-scale simulator
$W_{12}$
$W_{23}$
$V_1$
$V_2$
$V_3$
$(\bar{y}_i, \bar{s}_i)$
(c) Equivalent surrogate $\tilde{G}$
explicit edges $E_{ij}$ between bridge nodes
rates $\lambda_{ij} = \beta W_{ij} \bar{y}_i \bar{s}_j$; uniform import
weight $W_{ij}/(N_i N_j)$ on every pair